Koushik Swaminathan

Contact Information

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Education

D.Sc. Engineering Physics: Aalto University, Finland 2021–2026
Advisor: Dr. Sebastiano Peotta; Supervisor: Prof. Päivi Törmä
Quantum Dynamics Group, Dept. of Applied Physics
Thesis: Quasiparticle localization and ergodicity-breaking in flat-band lattice models

M.Sc. Physics: Ludwig-Maximilians-Universität München, Germany 2016–2019
Supervisor: Prof. Dr. Matthias Punk
Chair for Theoretical Solid State Physics, LMU Munich
Thesis: Conductivity in a Honeycomb Sachdev-Ye-Kitaev Lattice

B.Tech. Electronics & Communication Engineering: National Institute of Technology - Tiruchirappalli, India 2012–2016
Supervisors: Dr. P. Palanisamy, Dept. of Electronics and Communication Engineering;
Dr. Hemalatha Thiagarajan (Retd.), Dept. of Mathematics
Thesis: Study of quantum state tomography algorithms

Publications

- **K. Swaminathan**, A. Moustaj, J. Lado, S. Peotta, Parity-dependent double degeneracy and spectral statistics in the projected dice lattice, *Phys. Rev. B* **114**, L051108 (2026).
- M. Teeriaho, V.-V. Linho, **K. Swaminathan**, S. Peotta, Coexistence of ergodic and nonergodic behavior and level spacing statistics in a one-dimensional model of a flat band superconductor, *Phys. Rev. Research* **7**, 013318 (2025).
- **K. Swaminathan**, P. Tadros, S. Peotta, Signatures of many-body localization of quasiparticles in a flat band superconductor, *Phys. Rev. Research* **5**, 043215 (2023).
- V. Mantena, **K. Swaminathan**, M. Kumar, Magneto-oscillations in pentalayer graphene probed using Corbino geometry, *Article in preparation.*

Selected Presentations

Talks

- Parity-dependent spectral statistics in a flat-band lattice model, CMPCity 2026, London, United Kingdom Jul 2026
- Signatures of many-body localization of quasiparticles in a flat band superconductor, CDNBQS 2023, Évora, Portugal Oct 2023
- Multiple contributed talks at Physics Days (organized annually by the Finnish Physical Society) from 2023 to 2026

Posters

- Coexistence of ergodic and nonergodic behavior in a one-dimensional model of a flat band superconductor:
  · BEC 2025, Sant Feliu, Spain Sep 2025
  · SimQuDyn 2025, LMU Munich, Germany Apr 2025

- Signatures of many-body localization of quasiparticles in a flat band superconductor:
  · FINESS 2024, Gold Coast, Australia Sep 2024
  · QACTUS 2023, Barcelona, Spain Sep 2023
  · Quantum Many-Body Phenomena out of Equilibrium, ICTP Trieste, Italy Aug 2023

Research

Quasiparticle localization and ergodicity-breaking in flat-band lattice models 2022–

– Studied strongly correlated flat-band lattice models using exact diagonalization and symmetry-resolved spectral statistics.
– Derived exact many-body eigenstates of projected Hubbard Hamiltonians and showed coexistence of quasiparticle localization with pairing transport.
– Identified parity-dependent level statistics and exact double degeneracies in the projected dice-lattice model.

Magneto-oscillations in pentalayer graphene 2025–

– Developed Python-based tight-binding and continuum ($k \cdot p$) calculations for the band structure of multilayer graphene.
– Computed Landau fan diagrams and Shubnikov–de Haas oscillations for pentalayer graphene,and compared them with Corbino-geometry transport experiments performed at the Low Temperature Laboratory, Aalto University to identify signatures of Lifshitz transitions.

Conductivity in a honeycomb hyper-lattice with Sachdev-Ye-Kitaev dots 2018–2019

– Built a model of itinerant electrons coupled to Sachdev-Ye-Kitaev dots on a honeycomb lattice.
– Derived transport coefficients analytically and showed a finite-temperature crossover from a Dirac fermion regime to an incoherent strange-metal phase.

Teaching Experience

Teaching Assistant, Aalto University 2022–2025
Undergraduate: Classical Mechanics; *Graduate*: Low Temperature Physics – Nanoelectronics

Teaching Assistant, LMU Munich 2017–2019
Undergraduate: Classical Mechanics, Mathematical Methods for Physics, Theoretical Mechanics, Quantum Mechanics, Thermodynamics and Statistical Physics; *Graduate*: Advanced Statistical Physics

Course Material Support, LMU Munich 2018–2019
Advanced Statistical Physics; Proofreading assistance for *Mathematical Methods for Physicists* by A. Altland & J. von Delft

Funding

Travel Support
– CMPCity 2026, London, United Kingdom - £500
– BEC 2025, Sant Feliu, Spain - €1000
– CDNBQS 2023, Évora, Portugal - €500

Skills

Research areas: many-body physics, flat-band systems, random-matrix theory, quantum simulation
Computational methods: exact diagonalization, spectral statistics
Programming: Python, C, C++, Mathematica
Languages: English (native), Tamil (native), Hindi (basic), German (basic)

References

Dr. Sebastiano Peotta
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Dr. Manohar Kumar
Research Scientist, Low Temperature Laboratory, Dept. of Applied Physics, Aalto University
Email: manohar.kumar@aalto.fi

Prof. Dr. Sebastian Huber
Lecturer, Institute of Theoretical Physics, Department of Physics, ETH Zürich, Switzerland
Contact information provided on request.